

Lecture 8: Clustering

Lecturer: Prof. LI Yu

Scriber: ZHANG, Laibohan

Outline of Lecture 8:

- Recap from last lecture
- Hierarchical clustering
- Mahalanobis distance (Advanced topic)

Part 1. Recap from last lecture

A. What is clustering analysis?

Finding groups of objects such that the objects in a group will be similar(or related) to one another and different from(or unrelated to) the objects in other groups.

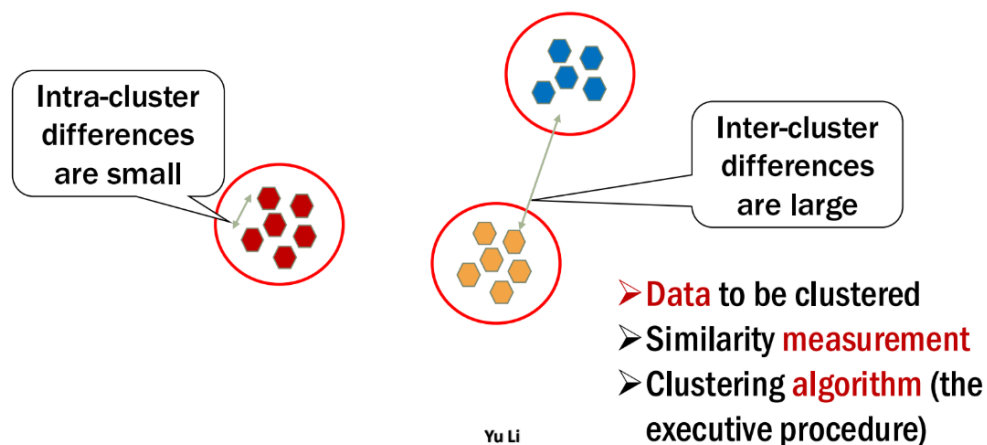


Figure 1. Definition of clustering analysis

During this procedure, those data with similar features will be merged together, which makes it easier for us to compare data from different groups and improves the efficiency and quality of data analysis.

B. Minkowski distance

It's a way to define the distance between two points in space, and here we use it as an index to compare the similarities/distance between different data point/cluster.

$$dist(p, q) = \left(\sum_{k=1}^m |p_k - q_k|^r \right)^{\frac{1}{r}}$$

Depending on various “r” values, the formula can have different structures and meaning.

1. $r = 1$: City block (Manhattan, taxicab, l_1 norm) distance.

A common example of this is the Hamming distance, which is just the number of bits that are different between two binary vectors.

2. $r = 2$: Euclidean distance.
3. $r \rightarrow \infty$: “supremum” distance.

This is the maximum difference between any component of the vectors.

Part 2. Hierarchical clustering

A. What is Hierarchical clustering?

Usually, in clustering analysis, we use similarity or dissimilarity (distance) to measure the differences (e.g. Cosine similarity, Correlation, Euclidean distance, Manhattan distance, Mahalanobis distance).

And after we calculate the similarities/dissimilarities, we need to do clustering for the data. Hierarchical clustering is a common way to cluster data: it analyzes the data layer by layer, and merges individual data to form small clusters; then those small clusters will be further merged to form huge clusters. Finally, it forms a hierarchical structure like The Tree of Life.

- Produces a set of nested clusters organized as a hierarchical tree;
- Can be visualized as a dendrogram (A tree-like diagram that records the sequences of merges);
- They may correspond to meaningful taxonomies (Gene clusters. phylogeny reconstruction, animal kingdom...).

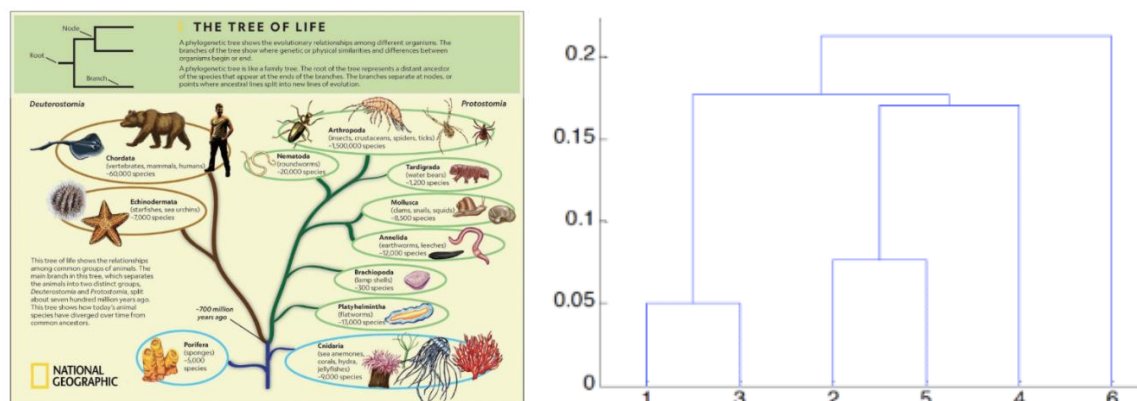


Figure 2. Hierarchical Clustering

B. Steps of hierarchical clustering

1. Compute the Similarity or Distance matrix

	p1	p2	p3	p4
p1	0	2.828	3.162	5.099
p2	2.828	0	1.414	3.162
p3	3.162	1.414	0	2
p4	5.099	3.162	2	0

Figure 3. Distance Matrix

2. Let each data point be a cluster
3. Merge the two closest clusters (Here: p2 and p3)
4. Update the similarity or distance matrix
5. Merge the two closest clusters
6. Update the similarity or distance matrix
-
0. Until only a single cluster remains

C. How to update the distance matrix after merging

Usually, there are many types of distance matrices, including Min distance, Max distance...And different distance matrices will have different method to update them.

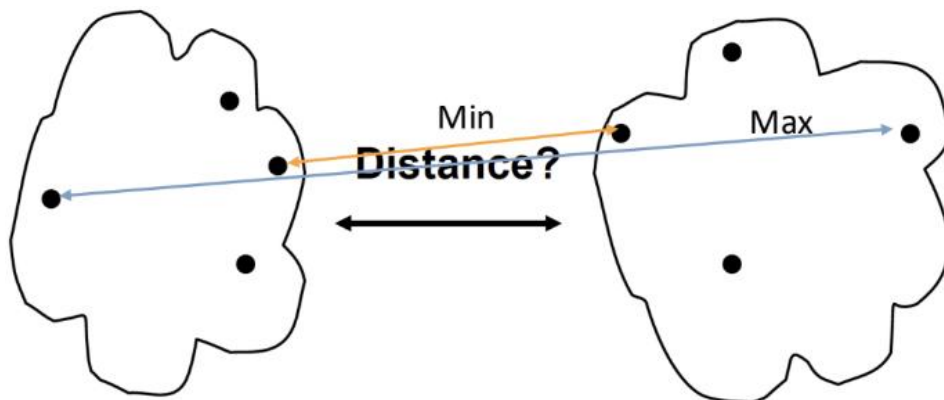


Figure 4. Update the distance matrices

- **Min distance matrix**

First, list all the data points in the 2 clusters, and then calculate the distance between the different data points of the 2 clusters. The pair with the minimum distance will represent the distance between these 2 clusters.

- **Max distance matrix**

First, list all the data points in the 2 clusters, and then calculate the distance between the different data points of the 2 clusters. The pair with the maximum distance will

represent the distance between these 2 clusters.

- Group average

Calculate the average data within a cluster and use it to represent this cluster; then compare the average data between different clusters to get the distance.

- Distance between centroids

Calculate the data coordinates of the centroids in the 2 clusters, and then calculate the distance between 2 centroids.

D. A running example

Gene	wt	mutant_1	mutant_2	mutant_3
At4g35770	1.5	3	3	1.5
At1g30720	4	7.5	7.5	5
At4g27450	1.5	1	1	1.5
At2g34930	10	25	23	15
At2g05540	1	1	2	1

Figure 5. A running example

1. Determine the Distance we should use

Because we want to find the similarity between different gene, we should use Correlation (linear correlation). The higher the correlation, the more similar the 2 genes.

$$\rho_{X,Y} = \text{corr}(X, Y) = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{E[(X - \mu_X)(Y - \mu_Y)]}{\sigma_X \sigma_Y}$$

2. Calculate the similarity matrix and let each data point be a cluster

	cluster 1	cluster 2	cluster 3	cluster 4	cluster 5
At4g35770	1				
At1g30720	0.9733	1			
At4g27450	-1	-0.9733	1		
At2g34930	0.9493	0.9909	-0.9493	1	
At2g05540	0.5774	0.562	-0.5774	0.4528	1

Figure 6. Similarity matrix

3. Merge the two closest clusters

	cluster 1	cluster 2	cluster 3	cluster 2	cluster 4
	At4g35770	At1g30720	At4g27450	At2g34930	At2g05540
At4g35770					
At1g30720	0.9733				
At4g27450	-1	-0.9733			
At2g34930	0.9493	0.9909	-0.9493		
At2g05540	0.5774	0.562	-0.5774	0.4528	

Figure 7. Merge cluster 1 & 2

4. Update the similarity matrix (**Tricky part**)

As the linear correlation represents the similarity between different clusters, so the higher correlation means higher similarity (closer).

To update the similarity matrix, after merge cluster 2 & 4, We need to re-examine the correlations between the new clusters 1, 2, 3 & 4. (the cluster 2 is the merged one). To compare merged cluster 2 with other clusters, we compare the 2 different data in the cluster 2 with the data of other clusters, and preserve the result with higher correlation (which means higher similarity). Then we refill the similarity matrix with new results.

	At4g35770	At1g30720	At4g27450	At2g34930	At2g05540
At4g35770					
At1g30720	0.9733				
At4g27450	-1	-0.9733			
At2g34930	0.9493	->0.9733	-0.9493		
At2g05540	0.5774	0.562	-0.5774	0.4528	

Figure 8. Update the similarity matrix

5. Repeat the above procedure until only a single cluster remains.

	At4g35770	At1g30720	At4g27450	At2g34930	At2g05540
At4g35770					
At1g30720					
At4g27450	-0.5774	-0.5774			
At2g34930			-0.5774		
At2g05540			-0.5774		

Figure 9. Final similarity matrix

The final result is a hierarchical structure, which looks like the Tree of Life.



Figure 10. Hierarchical structure

E. About programming

Scikit-learn

- Provide simple and efficient tools for predictive data analysis
- Accessible to everybody, and reusable in various contexts
- Built on NumPy, SciPy, and matplotlib
- Open source, commercially usable – BSD license

Scikit-learn

➤ <https://scikit-learn.org/stable/>

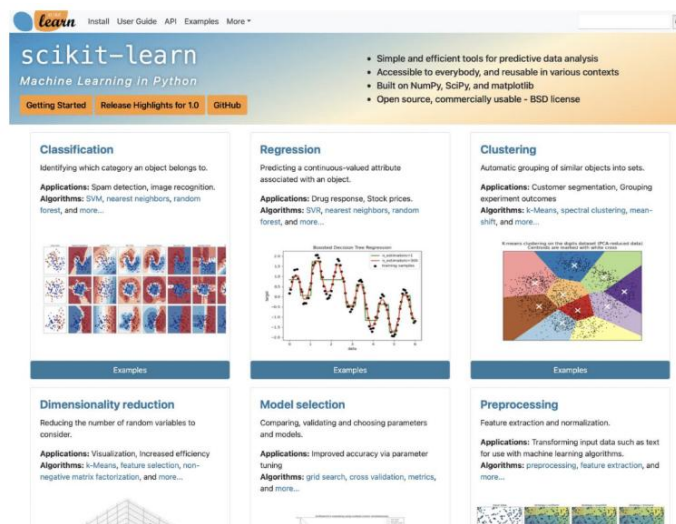


Figure 11. Scikit-learn website page

Part 3. Mahalanobis Distance (Advanced topic)

A. Definition of Mahalanobis Distance

It's a way to calculate the distance between different data point while considering the data distribution. For example, in the following graph: if we only consider the distance between 2 points, then A will be closer to B; but if we consider the data distribution, then A will be closer to C because both A and C are along the data distribution line.

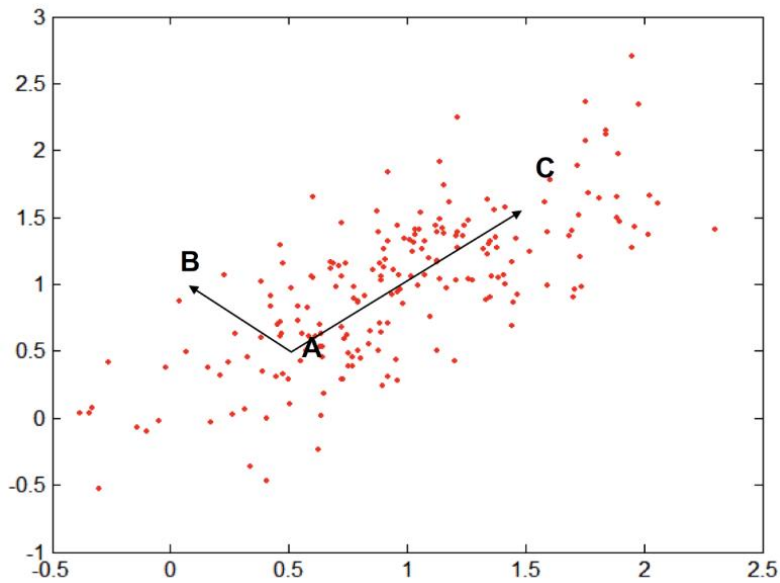


Figure 12. Example of Mahalanobis distance

B. The mathematical definition

The Mahalanobis distance can be derived from the following formula:

$$mahalanobis(p, q) = (p - q)^T \Sigma^{-1} (p - q)$$

Where the Σ denotes the covariance matrix.

Then there's a question: How to calculate the inverse of the covariance matrix?

We can derive it using mathematical solution:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

↑
determinant

$$\begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}^{-1} = \frac{1}{4 \times 6 - 7 \times 2} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix}$$

$$= \frac{1}{10} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{bmatrix}$$

Figure 13. How to get the inverse matrix

Now using the mathematical method, we can quantitatively calculate the distances between each point and compare them. According to the results, we can find that when calculating the Mahalanobis distance which considers the data distribution, the distance between A and C is closer than A and B ($4 < 5$).

Covariance matrix: **A: (0.5, 0.5)**

$$\Sigma = \begin{bmatrix} 0.3 & 0.2 \\ 0.2 & 0.3 \end{bmatrix}$$

B: (0, 1) Mahal(A,B) = 5

C: (1.5, 1.5) Mahal(A,C) = 4

Figure 14. Calculation of Mahalanobis distance

C. More examples

Suppose we have two quizzes, and the distributions of the grade are different.

Mahalanobis distance

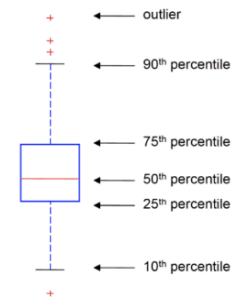
❖ Suppose we have two quizzes

➤ Quiz 1: std=10

- Student A-student B=1

➤ Quiz 2: std=1

- Student C-student D=1



❖ Should the two differences (A-B, C-D) be considered the same?

Figure 15. More examples

In this situation, the 2 differences can't be considered the same, because the variances of the results of the two quizzes are different; if we consider the distribution of the data, the difference between C and D is greater than that between A and B.

References

Li, Yu (2025). "Clustering".